Relevance is *not* enough: A Communication-Oriented Retrieval System for Consequential Scientific Question Answering

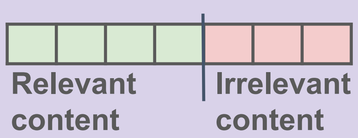
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Research Question: Can relevant context alone answer consequential scientific questions?

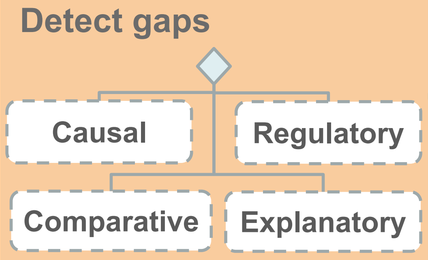
Gather context — *Two-stage retrieval audit: precision (prognostic) → sufficiency (diagnostic)* — *Draft answer* — *Recalibrate language*

Retriever
Find similar content
Search content topically similar to:
- Query
- Gaps

→ **Prognostic check**
Distill by relevance



→ **Diagnostic Check**
Detect gaps



→ **Generator**
Answer in context.
- Address the query
- Clarify uncertainties
- State remaining gaps

→ **Communicator**
Enhance accessibility
City Hall Rep. | Caring Neighbor
- Adjust empathy
- Raise readability
- Preserve facts and gaps

Method: Iterative retrieval targeting evidential sufficiency to generate complete answers

Collecting relevant context

Diagnosing evidentiary gaps across reasoning strategies

Causal
How does lead get into tap water?
- **Cause identified:** Where is lead hiding in the system?
- **Effect described:** How much lead gets in, and is it dangerous?
- **Causal pathway:** How does it leak into the water?
- **Temporal precedence:** Were lead pipes there before levels jumped?
- **Confounding factors:** What else adds lead to the water?
- **Counterfactual or control:** What about homes without lead pipes?

Caring Neighbor: It's completely natural to worry about our water, but lead mostly enters when older pipes—installed decades before levels spike—slowly corrode, leaching harmful lead into the flow. While indoor brass fixtures and old solder can also add lead, homes built without lead pipes remain largely protected from this primary hazard.

City Hall Rep.: Residents deserve clarity on public infrastructure risks, where water corrodes legacy service lines—laid long before modern bans—leaching dangerous lead into household supplies. Although household brass fittings and vintage solder can also introduce lead, properties with modern, lead-free plumbing consistently prevent this contamination.

Regulatory
What does it mean when the city's report say "non-detect" lead levels?
- **Concept or policy identified:** What does "non-detect" mean?
- **Purpose stated:** Why do reports use "non-detect"?
- **Mechanics explained:** How do they test for lead?
- **Compliance requirements:** What do cities have to report?
- **Changes or evolution:** Have the testing rules changed?
- **Implications for the person:** Does "non-detect" mean my water is safe?

Caring Neighbor: It's completely natural to wonder what this means for your family, but "non-detect" just means lab testing on stagnant tap water found lead levels lower than equipment can measure rather than claiming an inaccurate "zero." Even as federal testing rules grow stricter for city reports, we still encourage testing your water.

City Hall Rep.: "Non-detect" means certified lab testing of mandated stagnant samples measured lead below instrument detection limits rather than declaring an unverified "zero." Under updated EPA compliance regulations requiring stricter municipal monitoring, this result confirms full regulatory compliance.

Comparative
What is the difference between having 20 ppb and 5 ppb of lead?
- **Both comparators named:** What do 20 ppb and 5 ppb mean?
- **Shared metrics used:** What does "ppb" actually measure?
- **Reference or baseline present:** How do these compare to safe limits?
- **Structural differences**: Does how you test change the result?
- **Reason for the difference:** Why would two taps have different readings?

Caring Neighbor: It's completely natural to worry when seeing these numbers, where "ppb" (parts per billion) measures lead concentration—placing 5 ppb safely below federal action limits while 20 ppb exceeds them. Results vary between homes because differences in plumbing age, lead service lines, and testing methods change how much lead enters the sample.

City Hall Rep.: Parts per billion (ppb) measures lead concentration, meaning 5 ppb remains under municipal action levels while 20 ppb triggers regulatory intervention. Concentration variations across properties stem from specific plumbing composition, stagnation duration, and testing protocols.

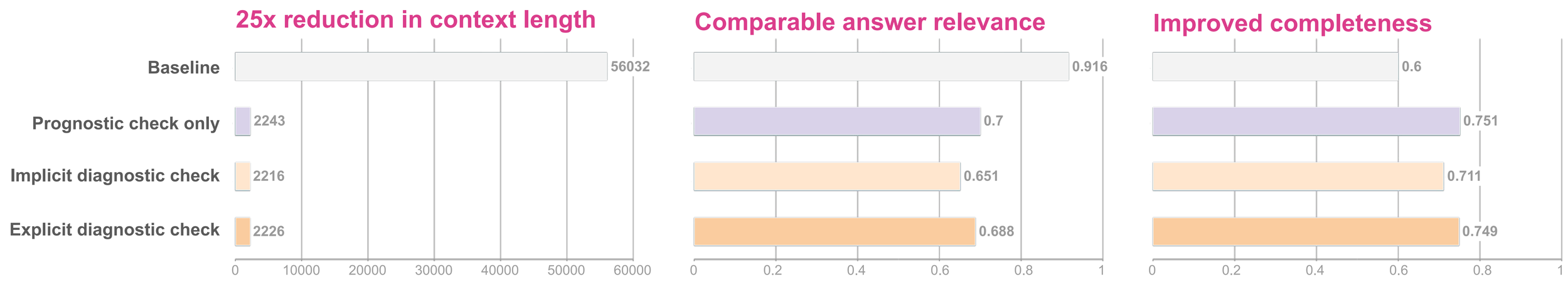
Explanatory
What does action level for lead mean?
- **Clear definition:** What does "action level" mean?
- **Scope and application:** Who does this rule apply to?

Caring Neighbor: The "action level" is simply a community safety trigger for public water systems rather than a safe health limit for an individual tap. This rule applies directly to municipal water utilities, requiring them to legally notify the public, fix water chemistry, and replace lead pipes if more than 10% of tested local homes exceed the limit.

City Hall Rep.: The "action level" is a federal regulatory threshold that triggers mandatory intervention rather than a health-based exposure limit. This standard applies to all public water utilities, legally municipality must implement corrosion controls, inform the public, and replace lead service lines if more than 10% of sampled taps exceed the threshold.

Constructing sufficient context

Evaluation: An order of magnitude less context, with higher human-rated completeness



Takeaway: Iterative gap analysis can efficiently enhance RAG response quality.

Findings
- Highly **relevant answers often miss critical context**, with standard RAG metrics explaining just 1% of human-rated completeness.
- Diagnosing context for evidentiary gaps and iteratively retrieving missing details **improved answer completeness** from 0.60 to 0.751.
- Despite iterative fetching, filtering irrelevant context per step **cut context length 25-fold** without sacrificing answer relevance.
- Assigning adaptive personas **boosted readability and empathy** without compromising factual content.

Next steps
- Queries: Evaluate gap-aware retrieval on open-ended, multi-step questions to measure true value under higher reasoning demands.
- Customize diagnostic question templates to extend the architecture beyond water quality into other high-stakes fields.
- Use hybrid evaluation workflows with fine-tuned LLM judges and human review.

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Selected References
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